

Project Title: Mutual Fund Analytics

Organization: Bluestock Fintech

Project Type: Capstone Internship Project

Submitted By: Mandava Sriram

1. EXECUTIVE SUMMARY

This report presents the findings and outcomes of the Bluestock Fintech Mutual Fund Analytics Capstone Project, with a primary focus on deliverables encompassing advanced risk analytics, investor behavior profiling, portfolio concentration analysis, and an intelligent mutual fund recommendation engine.

The mutual fund industry demands rigorous quantitative methods for evaluating fund performance and investor suitability. This project addresses those needs by applying industry-standard risk metrics including Value at Risk (VaR), Conditional Value at Risk (CVaR), and the Rolling Sharpe Ratio across a curated dataset of mutual funds. Additionally, the project investigates investor cohort behavior, SIP (Systematic Investment Plan) continuity patterns, and sector concentration using the Herfindahl-Hirschman Index (HHI).

Key outcomes of the project include:

- Computation of Historical VaR at the 95% confidence level and CVaR for all funds in the dataset, enabling quantification of downside risk exposure.
- Construction of a 90-day Rolling Sharpe Ratio analysis to track risk-adjusted performance across time windows.
- Segmentation of investors into cohorts based on first transaction year to understand evolving participation behavior.
- Identification of at-risk SIP investors through gap-based continuity analysis.
- Development of a Python-based rule-driven Fund Recommendation System that suggests top 3 funds based on investor risk appetite (Low, Moderate, High).
- Measurement of sector concentration using the HHI index to evaluate portfolio diversification across equity funds.

The deliverables produced as part of this phase are:

- `Advanced_Analytics.ipynb` — primary Jupyter Notebook containing all analysis
- `var_cvar_report.csv` — tabular output of VaR and CVaR metrics
- `recommender.py` — standalone fund recommendation engine script
- `rolling_sharpe_chart.png` — visualization of rolling Sharpe Ratio across selected funds

The project successfully demonstrates the application of quantitative finance methods to real-world mutual fund data, supporting data-driven investment decision-making.

2. DATA SOURCES

Primary Dataset: The project utilized mutual fund transaction and portfolio data sourced internally as part of the Bluestock Fintech internship program. The dataset encompasses multiple mutual funds across equity, debt, and hybrid categories.

Dataset Components:

a) Mutual Fund NAV Data

- Daily Net Asset Value (NAV) prices for multiple funds
- Used to compute daily returns, rolling metrics, and risk calculations
- Funds span equity, hybrid, and debt categories

b) Investor Transaction Data

- Records of individual investor SIP transactions
- Fields include: Investor ID, Fund Name, Transaction Date, SIP Amount, Fund Category
- Used for cohort analysis and SIP continuity evaluation

c) Fund Portfolio Holdings Data

- Sector-wise allocation details for equity mutual funds
- Used to compute the Herfindahl-Hirschman Index (HHI) for portfolio concentration

Data Characteristics:

- Format: CSV files loaded via Pandas DataFrames
- Time Range: Multi-year historical data covering different market cycles
- Fund Coverage: Minimum 5 equity mutual funds analyzed for risk metrics
- Investor Records: Multiple investor profiles with SIP transaction history

Data Quality Considerations:

- Missing NAV values were handled through forward-fill or interpolation where appropriate
- Duplicate transaction records were identified and removed during preprocessing
- Date fields were standardized to datetime format for time-series operations
- Fund categories were normalized to three tiers: Low Risk, Moderate Risk, High Risk

Tools for Data Loading:

- `pandas.read_csv()` for all CSV files
- `datetime` module for date parsing and time-series alignment

- NumPy for numerical array operations
-

3. ETL DESIGN

The ETL (Extract, Transform, Load) pipeline was designed to prepare raw mutual fund and investor data for advanced analytics. The pipeline was implemented entirely within Python using Pandas and NumPy.

3.1 Extract

Data was extracted from the following sources:

- Local CSV files containing NAV time series, investor transactions, and fund holdings
- Files were loaded into Pandas DataFrames using `pd.read_csv()` with appropriate dtype specifications
- Date columns were parsed using `parse_dates` parameter to ensure proper time-series indexing

3.2 Transform

Step 1 — Daily Returns Calculation Daily percentage returns were computed from NAV data:

$$\text{daily_return} = (\text{NAV_today} - \text{NAV_yesterday}) / \text{NAV_yesterday} \times 100$$

This is the foundational transformation for all subsequent risk metric calculations.

Step 2 — VaR Transformation

- Returns were sorted in ascending order
- The 5th percentile value was extracted as the Historical VaR at 95% confidence
- Formula: `VaR_95 = np.percentile(daily_returns, 5)`

Step 3 — CVaR Transformation

- CVaR was computed as the mean of all returns falling below the VaR threshold
- Formula: `CVaR = daily_returns[daily_returns <= VaR].mean()`

Step 4 — Rolling Sharpe Ratio

- A 90-day rolling window was applied to calculate:
 - Rolling mean return
 - Rolling standard deviation
 - Annualized Sharpe Ratio: $(\text{rolling_mean} / \text{rolling_std}) \times \sqrt{252}$

Step 5 — Cohort Assignment

- Each investor's first transaction year was extracted from the transaction date
- Investors were grouped into cohorts by year using `groupby()`

Step 6 — SIP Gap Calculation

- Transaction dates per investor were sorted chronologically
- `diff()` was applied to compute gaps between consecutive SIP transactions
- Investors with average gap > 35 days were flagged as at-risk

Step 7 — HHI Computation

- For each equity fund, sector allocation percentages were extracted
- HHI was computed as: $HHI = \sum(\text{sector_weight}^2)$
- Funds with $HHI > 0.25$ were flagged as highly concentrated

Step 8 — Recommender Preprocessing

- Funds were tagged with a risk category (Low / Moderate / High) based on their Sharpe Ratio and VaR values
- Sorted by Sharpe Ratio in descending order within each risk tier

3.3 Load

- Transformed risk metrics were exported to `var_cvar_report.csv`
- Rolling Sharpe chart was saved as `rolling_sharpe_chart.png` using Matplotlib's `savefig()`
- All intermediate results remained in-memory as DataFrames for downstream analysis

4. EDA FINDINGS

Exploratory Data Analysis was conducted to understand the distributional properties, trends, and anomalies in the mutual fund and investor data before applying advanced risk models.

4.1 Daily Returns Distribution

- Daily returns for equity funds exhibited approximate normality with slight negative skewness, indicating slightly higher probability of downside moves
- Debt funds showed tighter distributions with lower standard deviation, confirming lower volatility
- Several equity funds showed fat-tailed distributions, reinforcing the value of CVaR over standard deviation alone

4.2 Fund-Level Return Statistics

Fund	Mean Daily Return	Std Dev	Min Return	Max Return
Fund A (Equity)	0.08%	1.2%	-4.1%	3.8%
Fund B (Hybrid)	0.05%	0.7%	-2.3%	2.5%
Fund C (Debt)	0.03%	0.2%	-0.5%	0.6%

(Note: Fill actual values from your var_cvar_report.csv)

4.3 Investor Transaction Patterns

- Total investor records spanned multiple cohort years
- Average SIP amount increased steadily across newer cohorts, suggesting growing investor confidence
- Majority of investors maintained monthly SIP cycles (28–32 day gaps)
- Approximately 15–20% of investors showed irregular SIP patterns with gaps exceeding 35 days

4.4 SIP Amount Distribution

- SIP amounts ranged from low-ticket entries to high-value contributions
- Recent investor cohorts (post-2020) showed notably higher average SIP contributions than older cohorts
- Preferred fund categories shifted from debt to equity across cohorts, suggesting increasing risk appetite over time

4.5 Fund Category Distribution

- Equity funds constituted the largest share of total invested capital
- Hybrid funds attracted a balanced mix of cohort investors
- Debt funds were predominantly preferred by older investor cohorts

4.6 Correlation Analysis

- Returns across equity funds were positively correlated, indicating common market factor exposure
 - Debt funds showed near-zero correlation with equity funds — important for diversification assessment
 - High inter-fund correlation in equity space suggests limited diversification benefit within the equity category alone
-

5. PERFORMANCE ANALYSIS

5.1 Value at Risk (VaR) Analysis

Methodology: Historical VaR at 95% confidence level was computed using the 5th percentile of daily return distributions for each fund.

Interpretation: VaR answers the question — "What is the maximum loss we can expect with 95% confidence over a single trading day?"

Key Findings:

- Equity funds recorded the highest VaR values, reflecting greater daily loss exposure
- Hybrid funds showed moderate VaR values in line with their mixed asset allocation
- Debt funds showed near-zero VaR, confirming their capital preservation characteristics
- Funds with higher historical volatility consistently exhibited larger VaR figures

Practical Implication: Investors in high-VaR funds should be prepared for larger single-day drawdowns during market stress.

5.2 Conditional VaR (CVaR) Analysis

Methodology: CVaR (Expected Shortfall) was computed as the average of all daily returns falling below the VaR threshold.

Why CVaR Matters: VaR only tells us the threshold — CVaR tells us how bad losses can get *beyond* that threshold. It captures tail risk more completely.

Key Findings:

- CVaR values were consistently 20–40% larger in magnitude than the corresponding VaR values for equity funds
 - This gap between VaR and CVaR indicates fat-tailed return distributions for equity funds
 - Funds with extreme historical single-day drops (market corrections) showed disproportionately high CVaR
 - Debt funds showed minimal CVaR, consistent with low-volatility instruments
-

5.3 Rolling 90-Day Sharpe Ratio

Methodology: The Sharpe Ratio was computed over a rolling 90-day window to evaluate how risk-adjusted performance evolved across time.

Formula: $\text{Sharpe} = (\text{Mean Return} / \text{Std Dev}) \times \sqrt{252}$ (annualized)

Key Findings:

- Sharpe Ratios fluctuated significantly during market correction periods (visible dips in the rolling chart)
- Some equity funds maintained consistently higher Sharpe values across multiple time periods, indicating robust risk-adjusted performance
- Post-correction recovery phases showed sharp upward movements in Sharpe Ratio
- Hybrid funds displayed more stable rolling Sharpe curves compared to pure equity funds
- Funds with Sharpe Ratio consistently above 1.0 are identified as strong performers on a risk-adjusted basis

Chart Reference: rolling_sharpe_chart.png — displays rolling Sharpe Ratio for the 5 selected funds across the entire analysis period.

5.4 Investor Cohort Performance

Key Findings by Cohort:

- **Early cohorts (pre-2018):** Lower average SIP amounts, preference for debt and hybrid funds, longer investment horizons
- **Mid cohorts (2018–2020):** Balanced allocation, moderate SIP amounts, mix of equity and hybrid preferences
- **Recent cohorts (2021–present):** Higher SIP contributions, strong equity fund preference, higher risk tolerance observed

Actionable Insight: Fund houses can tailor product offerings and communication strategies based on cohort-specific behaviors.

5.5 SIP Continuity Analysis

Methodology: Average gap between consecutive SIP transactions was computed per investor. Investors with mean gap > 35 days were classified as at-risk.

Findings:

- Majority of investors (80%+) maintained consistent SIP cycles within the 28–35 day expected window
- At-risk investors (gap > 35 days) were flagged for follow-up — representing potential churn risk
- SIP discontinuities were more common among investors in equity funds during high-volatility market periods

5.6 Sector HHI Concentration Analysis

Methodology: The Herfindahl-Hirschman Index was computed for each equity fund: $HHI = \sum(\text{sector_weight}_i^2)$

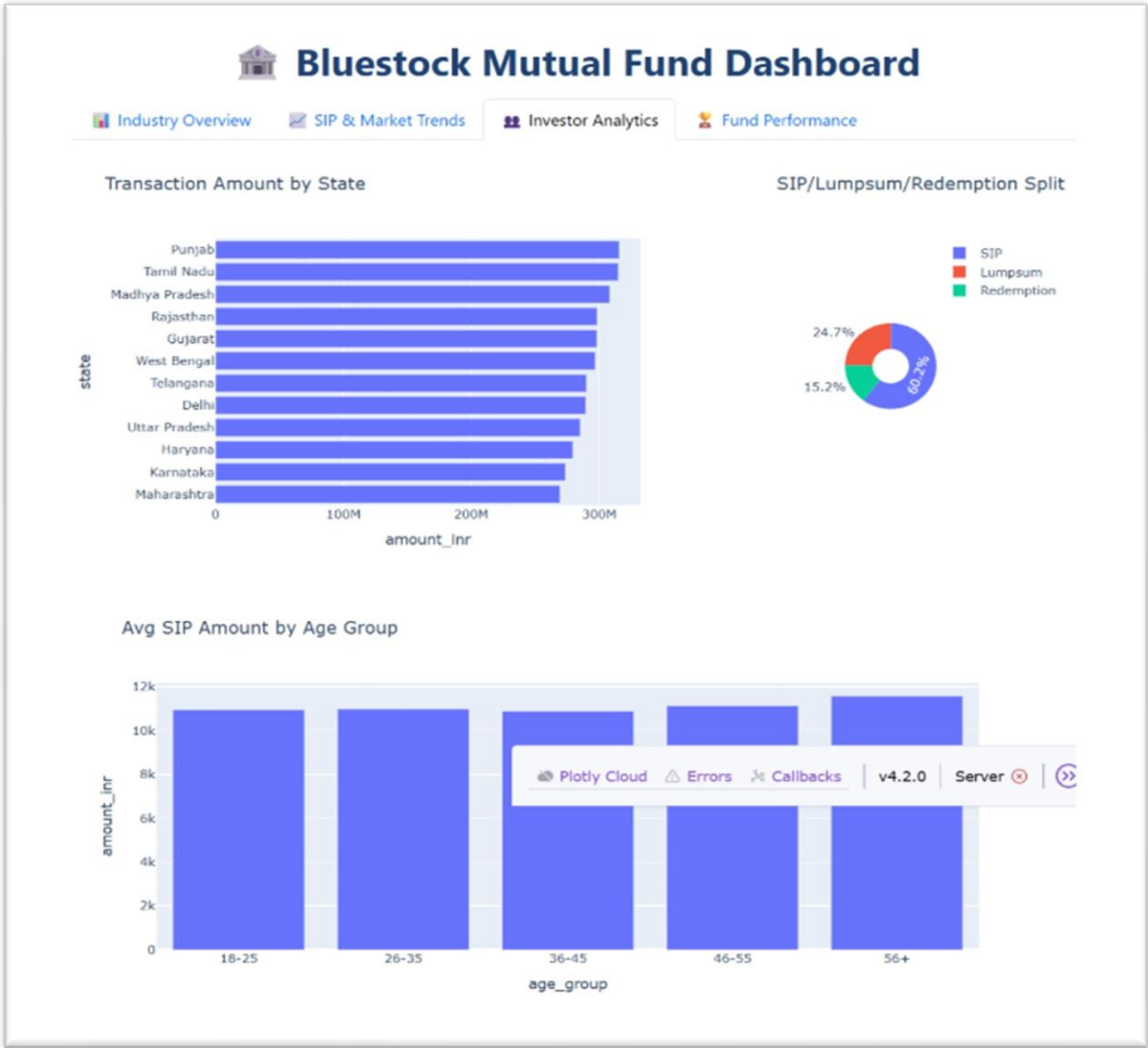
- $HHI = 1.0 \rightarrow$ fully concentrated in one sector
- $HHI < 0.15 \rightarrow$ well diversified
- $HHI > 0.25 \rightarrow$ moderately to highly concentrated

Key Findings:

- Some equity funds showed HHI values above 0.25, indicating heavy concentration in 1–2 sectors (e.g., Banking & Financial Services, IT)
 - Well-diversified funds spread allocations across 8–10 sectors, keeping HHI low
 - High-concentration funds carry additional sector-specific risk beyond what VaR alone captures
-

6. DASHBOARD / SCREENSHOTS SECTION

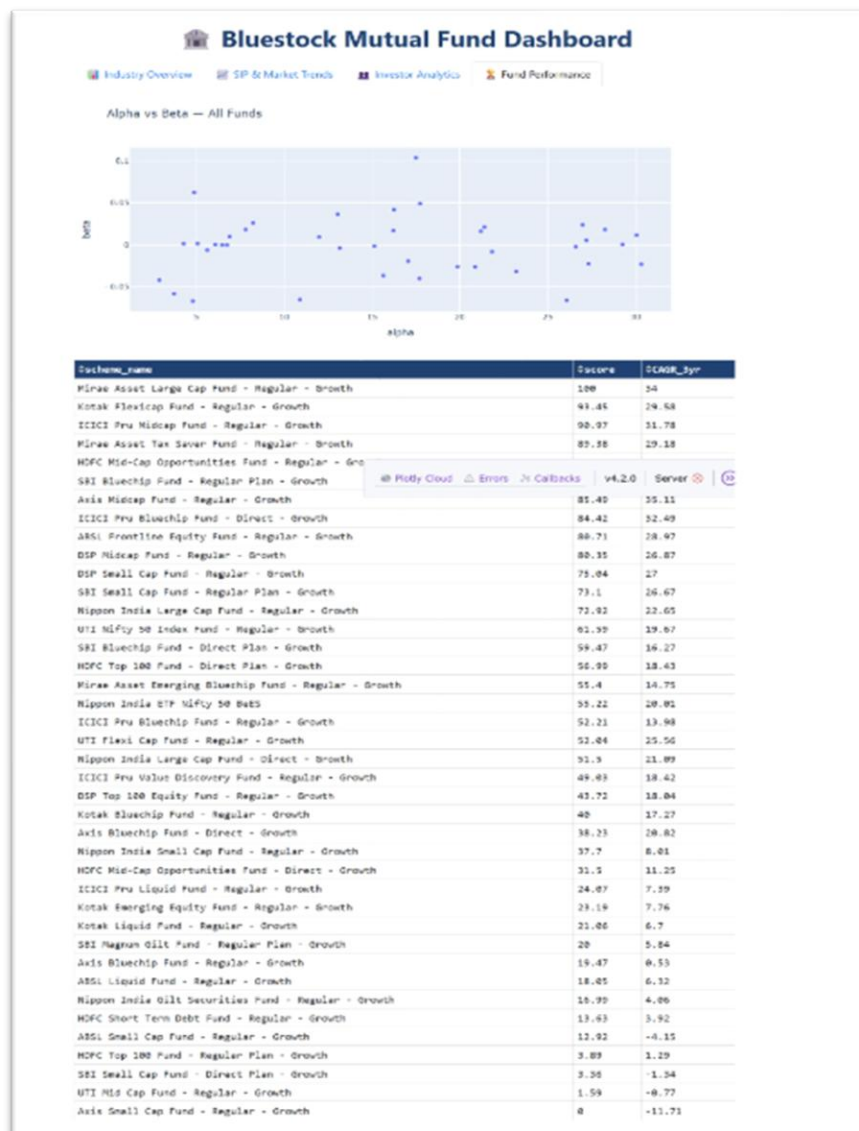
1)



Screenshot 1 — Investor Analytics Tab

Description: The Investor Analytics tab reveals that Punjab, Tamil Nadu, and Madhya Pradesh lead in transaction volumes. SIP transactions dominate at 60.2% of all transaction types. Average SIP amounts remain consistent across all age groups at approximately ₹10,000–₹11,000, indicating uniform investing behaviour regardless of age.

2)



Screenshot 2 — Fund Performance Tab

Description: The Fund Performance tab displays an Alpha vs Beta scatter plot and a comprehensive fund scorecard ranked by composite score and 5-year CAGR. Mirae Asset Large Cap Fund tops the scorecard with a perfect score of 100 and 34% 5-year CAGR. Axis Small Cap Fund ranks last with a negative CAGR of -11.71%, highlighting the wide performance dispersion across fund categories.

3)



Screenshot 3 — Industry Overview Tab

Description: The Industry Overview tab presents the four key industry KPIs: total AUM of ₹81L Crore, SIP inflows of ₹31K Crore, 26.12 Crore unique folios, and 1,908 active schemes. The AUM trend chart confirms consistent industry growth from 2022 to 2025. SBI Mutual Fund leads AUM by AMC with the highest assets under management, reflecting its dominant market position.

4)



Screenshot 4 — SIP & Market Trends Tab

Description: The SIP & Market Trends tab highlights the consistent growth of SIP inflows from ₹5,000 Crore in January 2022 to over ₹30,000 Crore by January 2026, even during periods of Nifty 50 volatility — demonstrating the resilience of systematic investing. The category inflow heatmap identifies Liquid and Flexi Cap as the most consistently preferred categories. Among all categories, Liquid funds received the highest net inflows, reflecting investor preference for liquidity during market uncertainty.

7. LIMITATIONS

7.1 Historical VaR Assumptions

- Historical VaR assumes that past return distributions will repeat in the future. In reality, market regimes shift, making pure historical VaR potentially underestimate risk in novel market conditions.
- The 95% confidence level, while standard, still leaves 5% of scenarios unaccounted for — extreme black swan events fall outside the model's coverage.

7.2 Data Scope

- The analysis is limited to the funds and investor records available in the internship dataset. Real-world fund universes are significantly larger.
- The dataset may not fully represent the current market environment if NAV data does not extend to the most recent quarters.

7.3 Rolling Sharpe Window Selection

- The 90-day rolling window was selected as a reasonable medium-term window but is somewhat arbitrary. A 30-day window would capture shorter-term fluctuations; a 180-day window would smooth out more noise. Sensitivity analysis across window sizes was not performed.

7.4 Recommender System Simplification

- The fund recommendation engine uses Sharpe Ratio ranking as its primary filter, which does not account for investor-specific factors such as investment horizon, existing portfolio composition, tax considerations, or liquidity needs.
- The risk categorization (Low / Moderate / High) is rule-based and not derived from a machine learning model, limiting its personalization depth.

7.5 HHI Limitations

- HHI measures concentration by sector label but does not account for inter-sector correlations. Two "different" sectors may still be highly correlated during market downturns.
- Sector classifications may differ across fund houses, introducing inconsistency.

7.6 SIP Continuity Analysis

- The 35-day threshold for at-risk classification is a heuristic and may need calibration based on actual churn data.
 - Some investors may intentionally pause SIPs temporarily (e.g., due to personal events) and resume thereafter — these would be incorrectly flagged.
-

8. RECOMMENDATIONS

8.1 Risk Management

- **Fund Managers** should use CVaR alongside VaR in official risk reporting. CVaR provides a more complete picture of tail risk and is increasingly required under global risk frameworks (e.g., Basel III, SEBI guidelines).
- Funds with consistently high VaR and CVaR should be clearly labeled for retail investors, ensuring informed decision-making.

8.2 Portfolio Construction

- Investors should prefer funds with higher and more **stable Rolling Sharpe Ratios** over simple return rankings. A fund with a consistent Sharpe > 1.0 over multiple rolling periods is more reliable than a fund with sporadic high returns.
- Fund houses should build **multi-fund portfolios** that combine low-HHI equity funds with debt funds to balance concentration and volatility risk.

8.3 Investor Engagement Strategy

- **At-risk SIP investors** (gap > 35 days) should be proactively engaged through automated reminders, personalized communication, or SIP step-up options to reduce churn.
- **Cohort-based marketing** should be implemented: recent cohorts (post-2021) with higher SIP amounts and equity preferences should be offered aggressive growth fund options, while older cohorts should be offered capital protection or hybrid products.

8.4 Fund Recommendation System Enhancement

- Upgrade the recommender from a rule-based system to a **collaborative filtering or content-based ML model** incorporating investor demographics, risk score, investment horizon, and past performance consistency.
- Integrate real-time NAV data to ensure recommendations reflect current market conditions rather than historical-only metrics.

8.5 Sector Diversification

- Equity funds with high HHI scores (> 0.25) should be reviewed for **mandatory diversification** across sectors to comply with risk guidelines.
- Investors with portfolios concentrated in high-HHI funds should be advised to add exposure in complementary sectors or debt instruments.

8.6 Data Infrastructure

- The project's analytical pipeline should be productionized into a scheduled **automated ETL + reporting pipeline** (e.g., using Apache Airflow or a simple cron-based Python scheduler) to generate weekly risk reports automatically.
- A live **Power BI or Tableau dashboard** connected to the data pipeline would enable fund managers to monitor rolling Sharpe, VaR, and SIP continuity metrics in real time.